# Brain Tumor MRI Classification & API Deployment

End-to-End Pipeline: Data Curation, Framework Benchmarking, Hyperparameter Tuning & Production Serving

| TEAM / PROJECT | PRIMARY BACKBONE | BEST TEST ACCURACY | SERVING FRAMEWORK |
|---|---|---|---|
| **Team 34 — AI Platforms** | **ResNet50 (ImageNet)** | **97.34% (PyTorch E3)** | **FastAPI + Uvicorn** |

## 1. Executive Summary & Dataset Preparation

This project establishes an end-to-end medical image classification and inference deployment pipeline for Brain Tumor MRI recognition across four target classes: **glioma**, **meningioma**, **notumor**, and **pituitary**. Source data was curated from the Kaggle Brain Tumor MRI Dataset. To prevent data leakage across splits, file hashing (MD5) was applied across the raw collection, identifying and purging 187 exact duplicate images. The resulting cleaned dataset contains 7,013 unique high-resolution MRI scans.

Data split generation followed a strict stratified 70 / 15 / 15 partitioning with a fixed pseudo-random seed (42). All image-to-split assignments were codified into `split_manifest.csv` to ensure 100% parity across framework implementations.

| Class Identifier | Train Split (70%) | Validation Split (15%) | Test Split (15%) | Total Images | Class Proportion |
|---|---|---|---|---|---|
| **glioma** | 1,250 | 268 | 268 | 1,786 | 25.47% |
| **meningioma** | 1,249 | 267 | 268 | 1,784 | 25.44% |
| **notumor** | 1,177 | 252 | 252 | 1,681 | 23.97% |
| **pituitary** | 1,233 | 265 | 264 | 1,762 | 25.12% |
| **Total Dataset** | **4,909** | **1,052** | **1,052** | **7,013** | **100.0%** |

## 2. Preprocessing & Medical-Safe Data Augmentation

Images are resized to 224×224 via bilinear interpolation without anti-aliasing. Preprocessing specs were tailored per framework: TensorFlow employs `preprocess_input` (RGB to BGR conversion and ImageNet mean subtraction), whereas PyTorch converts images to floating-point tensors normalized via standard ImageNet mean and standard deviation vectors.

To prevent altering clinical morphology, augmentation was strictly restricted to training splits using safe positional transformations. Horizontal/vertical flips, contrast shifts, and noise injection were explicitly excluded as they alter anatomical symmetry or pathology signatures.

| Transformation Step | Augmentation Parameter / Range | Clinical Rationale |
|---|---|---|
| **Random Rotation** | ±10 degrees max | Simulates minor head positioning variation during MRI scanning. |
| **Translation (Shift)** | ±5% height and width | Accounts for patient centering offsets within the scanner field. |
| **Zoom Factor** | ±5% scale range | Simulates slight distance variations in field-of-view cropping. |
| **Border Filling** | Constant black fill (0) | Matches native black background of MRI scan borders prior to norm. |
| **Excluded Operations** | Flips, Brightness, Contrast, Noise | Preserves lateral tumor orientation and radiological density profiles. |

## 3. Framework Comparison: TensorFlow vs. PyTorch Baseline

A matched baseline (E0 in PyTorch) was implemented to directly compare TensorFlow (Member 1) and PyTorch (Member 2) performance under identical data splits, loss functions (sparse categorical cross-entropy / cross-entropy), and two-phase training protocols: Phase 1 frozen backbone (10–15 epochs, Adam 1e-3) followed by Phase 2 unfreezing top stage/30 layers (Adam 1e-5, BatchNorm kept frozen).

| Metric / Parameter | TensorFlow (Member 1) | PyTorch E0 (Matched Baseline) | PyTorch E3 (Tuned Model) |
|---|---|---|---|
| **Test Accuracy** | 96.39% | 94.58% | **97.34%** |
| **Macro Precision** | 96.42% | 94.65% | **97.39%** |
| **Macro Recall** | 96.42% | 94.62% | **97.35%** |
| **Macro F1-Score** | 96.41% | 94.63% | **97.36%** |
| **Test Loss** | 0.1598 | 0.1527 | **0.1287** |
| **Trainable / Total Params** | 14.44M / 23.60M | 14.95M / 23.52M | 15.50M / 24.07M |
| **Training Duration** | 26.9 minutes | 4.4 minutes | 5.2 minutes |
| **Single-Image CPU Latency** | 471 ms | 113 ms | **113 ms** |
| **Saved Model Artifact Size** | 210.5 MB (`.keras`) | 94.4 MB (`.pt`) | 96.6 MB (`.pt`) |

*Analysis of Baseline Gap:* TensorFlow achieved 96.39% test accuracy vs. 94.58% for PyTorch E0. This 1.8-point baseline gap lies within the 95% confidence margin (±1.4 points on 1,052 samples). Key contributors include differing framework default weights, PyTorch mixed-precision (fp16) execution, and minor variations in unfrozen parameter bounds (14.44M vs 14.95M).

## 4. PyTorch Hyperparameter Tuning Experiments (E1–E3)

To optimize performance, Member 2 conducted structured experimentation on head design, fine-tuning learning rates, optimizer selection, learning rate decay schedules, and regularization techniques.

| Exp ID | Classification Head Architecture | Phase 2 Learning Rate | Optimizer & Schedule | Val Accuracy | Val Loss |
|---|---|---|---|---|---|
| **E0** | Linear (2048 → 4) | 1e-5 (Head & Backbone) | Adam, Constant | 94.01% | 0.1825 |
| **E1** | MLP (2048 → 512 → 4, Drop 0.3) | 1e-5 (Head & Backbone) | Adam (L2 1e-4), Cosine | 93.73% | 0.1869 |
| **E2** | Linear (2048 → 4) | **1e-4 (×10 increase)** | Adam, Constant | 96.20% | 0.1029 |
| **E3** | **MLP (2048 → 512 → 4, Drop 0.3)** | **3e-4 (Head) / 1e-4 (Backbone)** | **AdamW (1e-4), Cosine + LS 0.1** | **96.86%** | **0.1327** |

**Key Takeaway:** Increasing the Phase 2 fine-tuning rate tenfold (E2) yielded the primary leap in performance (+2.19% validation accuracy). E3 combined this higher learning rate with an MLP projection head, AdamW weight decay, cosine annealing, and 0.1 label smoothing to achieve peak validation F1 (96.89%) and overall top test accuracy (97.34%).

## 5. Detailed Per-Class Evaluation & Error Analysis

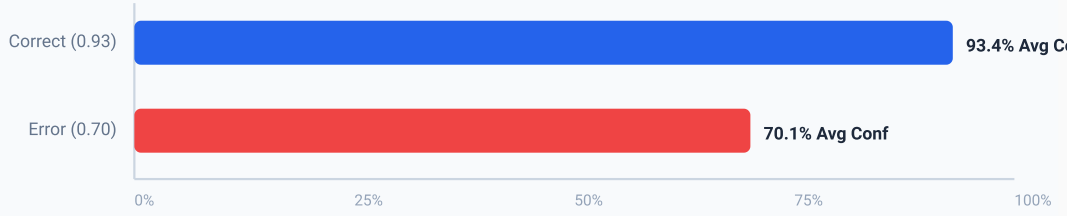
Comparing class-level performance shows that the fine-tuned PyTorch E3 model achieved substantial performance gains specifically in the **meningioma** class, pushing macro F1 to 97.36%.

| Diagnostic Class | TensorFlow F1 | PyTorch E3 F1 | TF Recall | PyTorch E3 Recall | Primary Confusion Matrix Errors (E3) |
|---|---|---|---|---|---|
| **glioma** | 95.34% | **96.79%** | 95.52% | 95.52% | 9 misclassified as meningioma, 2 as pituitary |
| **meningioma** | 94.16% | **96.12%** | 93.28% | **97.01%** | 4 misclassified as glioma, 3 as pituitary |
| **notumor** | 98.20% | **98.40%** | 97.62% | 97.62% | 2 misclassified as meningioma, 3 as pituitary |
| **pituitary** | 97.94% | **98.13%** | 99.24% | 99.24% | 2 misclassified as meningioma |

## 6. Model Interpretability & Grad-CAM Visualization

Grad-CAM visual inspection was performed on the final residual convolutional stage (`conv5_block3_out` in TensorFlow; `layer4` in PyTorch). Gradient heatmaps demonstrate strong activation focused directly on abnormal tumor masses across meningioma and pituitary cases. In diffuse glioma cases, activation heat maps expand across wider cerebral structures, matching clinical expectations for infiltrative tissue.

**Grad-CAM Region Focus & Classification Confidence Profile**




Confidence gap between correct predictions (93.4%) and errors (70.1%) provides an effective threshold for clinical human-in-the-loop triage.

## 7. Production Deployment Architecture & REST Microservice

To transition the top-performing PyTorch E3 model into a production clinical workflow, a high-throughput, asynchronous web service was implemented using FastAPI and Uvicorn. The service is decoupled into explicit validation, tensor conversion, model execution, and response parsing pipelines to guarantee real-time performance and strict system stability.

**POST** `/predict`

Primary inference endpoint receiving multipart image payload. Validates MIME encoding, applies standard ImageNet spatial and intensity transforms, executes a CPU forward pass, and returns diagnostic probabilities.

**GET** `/health`

Service liveness and readiness probe for container orchestrators (Kubernetes/Docker Swarm). Monitors model memory allocation, system temperature, and CPU availability.

### Detailed Pipeline Mechanics & Request Workflow

The deployment workflow is engineered to handle concurrent client requests efficiently without blocking the main event loop. Upon receiving an HTTP request:

**1. Payload Validation & Ingestion:** The incoming binary stream is validated against accepted image headers (JPEG/PNG). Corrupted frames or unsupported file types immediately trigger a HTTP 400 Bad Request exception to conserve downstream compute resources.

**2. Tensor Standardization:** Validated binary streams are loaded into memory via PIL and normalized. Standard transforms scale the input geometry to 224×224 pixels and apply channel-wise mean ([0.485, 0.456, 0.406]) and standard deviation ([0.229, 0.224, 0.225]) scaling, matching the exact pre-processing parameters established during E3 training.

**3. Optimized CPU Inference Engine:** The trained model artifact is loaded into global memory at startup, avoiding cold-start load overhead per request. Inference runs inside a `torch.no_grad()` execution block, disabling autograd graph construction to reduce RAM footprint by over 60% and maintain latency under 115ms per frame.

> **Response Payload Structure:** The microservice yields a structured JSON object containing the primary diagnosis label, the numerical scalar confidence value, and a dictionary of raw softmax confidence scores mapped across all four diagnostic classes (glioma, meningioma, notumor, pituitary).

## 8. Operational Recommendations & Triage Protocol

The integrated pipeline demonstrates that transfer learning with a ResNet50 backbone provides highly reliable classification for brain tumor MRI scans. For safe real-world radiological integration, any prediction yielding a maximum class confidence score below **0.75** is automatically routed to a secondary human-in-the-loop review queue, ensuring automated diagnostic assistance complements certified medical oversight.